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SEAT No. :

P71

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APR-17/BE/Insem.-81
B.E. (Computer Engineering) (Semester - II)
HIGH PERFORMANCE COMPUTING
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer any three questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Assume suitable data if necessary.*

Q1) a) What are applications of Parallel Computing? [4]

b) Explain basic working principal of VLIW processor. [6]

OR

Q2) a) Explain SIMD, MIMD and SIMT architecture. [6]

b) Write a short note on Dataflow Model. [4]

Q3) a) Explain Granularity, Concurrency, and Dependency Graph. [5]

b) What are the characteristics of Tasks and Interactions. [5]

OR

Q4) a) Explain Randomized block distribution and hierarchical mappings. [6]

b) Write a note on IBM Cell Broadband Engine (CBE). [4]

Q5) a) What are principles of Message Passing Programming. [4]

b) Write note on : Topologies and Embedding. [6]

OR

Q6) a) Explain Non-Blocking communications using MPI. [4]

b) Write short note on Blocking Message Passing Operations. [6]

